

CHUANGXI CHEN (陈创禧)

+86 150 1724 2417 | chuangxi031220@outlook.com

EDUCATION

Beijing Institute of Technology, Zhuhai | Zhuhai, China

Bachelor of Engineering in Electronic Science and Technology

2022/09 – 2026/07

- GPA: 3.51 / 5.0 | GPA Ranking: 6 / 33
- Selected Honors: First-Class Student Scholarship (2024); "Excellent Student" Honor (Consecutive years).
- Core Coursework: Mathematics: Advanced Mathematics (95), Probability & Statistics (90), Linear Algebra (87).
- Core Engineering: Digital Image Processing (Excellent), Embedded System Design (Excellent), Analog IC Design (Excellent), Communication Circuits (92), Semiconductor Physics (89).

PUBLICATIONS

- C. Chen, et al., "Dual-Domain Fractional Fourier Transformer for Underwater Image Degradation Removal," in *IEEE Signal Processing Letters (SPL)*, vol. 33, 2026. [First Author]
 - Impact Factor: 3.9 (JCR Q2) | DOI: 10.1109/LSP.2026.3660912
 - Significance: Proposed a novel dual-domain loss function combining Fractional Fourier Transform (FrFT) and Self-Attention to eliminate scattering distortion in underwater imagery.
- Y. Luo, Z. Chen, W. Qiu, C. Chen, et al., "WUE-Net: A Wavelet and EUCB Enhanced Framework for Accurate Building Change Detection in Remote Sensing Imagery," in *Proc. SPIE 13732, Fourth International Conference on Optics and Machine Vision (ICOMV 2025)*, 137320A, 2025. [Co-Author]
 - DOI: 10.1117/12.3072025 | Indexed in EI Compendex.
- C. Yang, H. Zhao, W. Qiu, Y. Deng, C. Chen, et al., "Building Change Detection Method Based on Multi-scale Feature Fusion and Uncertainty Awareness," in *Proc. SPIE 13690, International Conference on Advances in Computer Vision Research and Applications (ACVRA 2025)*, 136900N, 2025. [Co-Author]
 - DOI: 10.1117/12.3068588 | Indexed in EI Compendex.
- Patents: Two National Invention Patents Authorized (CN), related to image restoration algorithms.

RESEARCH EXPERIENCE

INTELLIGENT VISION RESEARCH LABORATORY (A KEY LABORATORY OF ZHUHAI) | RESEARCHER & PROJECT LEAD 2023/09 – Present
(Selected based on academic excellence; focused on underwater vision restoration and edge AI deployments)

- HDBFormer: RGB-D Semantic Segmentation in Complex Underwater Scenes | Project Lead
 - Performance (SOTA): Ranked 2nd on SUN-RGBD and 7th on NYUDepthv2 benchmarks on Papers with Code.
 - Methodology: Designed a Heterogeneous Dual-Branch Network to address modal discrepancies. Developed a Modal Information Interaction Module (MIIM) to fuse global (Transformer) and local (CNN) features.
 - Results: Achieved 87.6% mIoU on the SUIM dataset; compressed model to 0.7M parameters for real-time robotic inference.
- Underwater Polarization Camera System: Architecture & Embedded Deployment | System Architect
 - System Design: Engineered a custom polarization imaging system using a Jetson Nano (Algorithm) + STM32 (Control) master-slave architecture via Ethernet socket communication.
 - Hardware: Designed CNC aluminum alloy housing for waterproofing; performed thermal analysis to ensure heat dissipation.
 - Edge AI: Deployed a lightweight underwater enhancement algorithm using TensorRT, reducing latency by 40% and bandwidth occupancy by 40%.

PROFESSIONAL EXPERIENCE

CETC Potevio Technology Co., Ltd. (a listed company and a member of CETC) | Guangzhou, China

R&D Intern, PCB Design & Signal Integrity

2025/07 – 2025/09

- PCB Design: Utilized Cadence Allegro for multi-layer industrial control board layout. Focused on impedance

CHUANGXI CHEN (陈创禧)

+86 150 1724 2417 | chuangxi031220@outlook.com

matching for differential pairs and optimized power integrity (PI) for high-current modules.

- **EMC Optimization:** Conducted EMC testing; diagnosed signal interference issues and resolved them by optimizing ground plane partitioning and decoupling capacitor placement.
- **Verification:** Verified waveforms with oscilloscopes and performed thermal imaging analysis to ensure industrial-grade reliability.

Zhongyuan Technology (Guangdong) Co., Ltd. | Guangzhou, China

Algorithm Engineer Intern

2024/06 – 2024/08

- **Computer Vision:** Developed a defect detection system for power grid structures using YOLOv12. Curated datasets and tuned hyperparameters, achieving >95% detection accuracy in field trials.
- **Time-Series Forecasting:** Built an LSTM-based load forecasting model for the Southern Power Grid. Implemented data cleaning pipelines for missing industrial data and deployed the model to the production planning platform.

AWARDS & HONORS

- **Meritorious Winner (Top 7%),** Mathematical Contest in Modeling (MCM/ICM), USA (2024)
- **National Level Project,** Undergraduate Innovation & Entrepreneurship Training Program (2025)
- **Provincial 1st Prize,** 18th "Challenge Cup" Academic Science & Technology Competition (2025)
- **Rank 1 (School-wide),** 2025 College Students' Innovation Cultivation Project, Guangdong Science and Technology Innovation Strategy Special Fund (2025)
- **National 2nd Prize,** China Undergraduate Mathematical Contest in Modeling (CUMCM) (2023)

TECHNICAL SKILLS

- **Programming:** Python (PyTorch/TensorFlow), C/C++ (Embedded), MATLAB, Verilog.
- **Hardware Design:** Cadence Allegro, Altium Designer, STM32, FPGA (Xilinx), Signal Integrity Analysis.
- **AI & Deployment:** TensorRT, ONNX, YOLO Series, OpenCV, Linux (Ubuntu/ROS), Docker.
- **Certifications:** HarmonyOS Application Developer Senior Certification.